

Nodes Inventory

User Manual

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Nodes Inventory is an Android field tool for INOVA Quantum seismic nodes. It listens for nodes over Bluetooth Low Energy (BLE), lists them, shows them on a map, and runs a deploy-and-pickup accounting workflow so your crew can prove exactly which nodes went out into the field and which ones came back. Everything lives under three tabs at the bottom of the screen: Scan, Map, and Jobs.

Quick Start (101)

One full cycle, start to finish, in six steps.

1. Open the **Jobs** tab, type a job name, and press **New Job**.
2. With all the nodes together in their cradles or bags, press **Pre-Deploy**, then **SCAN**. Each node the phone hears is added to the job manifest. Press **Finish deploy** to save it.
3. Send the nodes out to the field.
4. Once the nodes are back and sitting together in their bags or cradles, open the job, press **Start Pickup**, then **SCAN**, then **Reconcile now**.
5. Watch the header count "R of M recovered" climb as each node in the bags or cradles is heard at strong signal and confirmed. Scan again if some are still missing.
6. Open **View Report**, press **Export CSV** to save and share the record, then press **Close job**.

TIP

Nodes broadcast every few seconds, not continuously, so give every scan a few seconds before you decide a node is not there.

The Three Tabs

Scan

A live Bluetooth scan. Every node in range shows up as a card with its label, signal, GPS, and battery. Use it to check a single node quickly, without opening a job.

Map

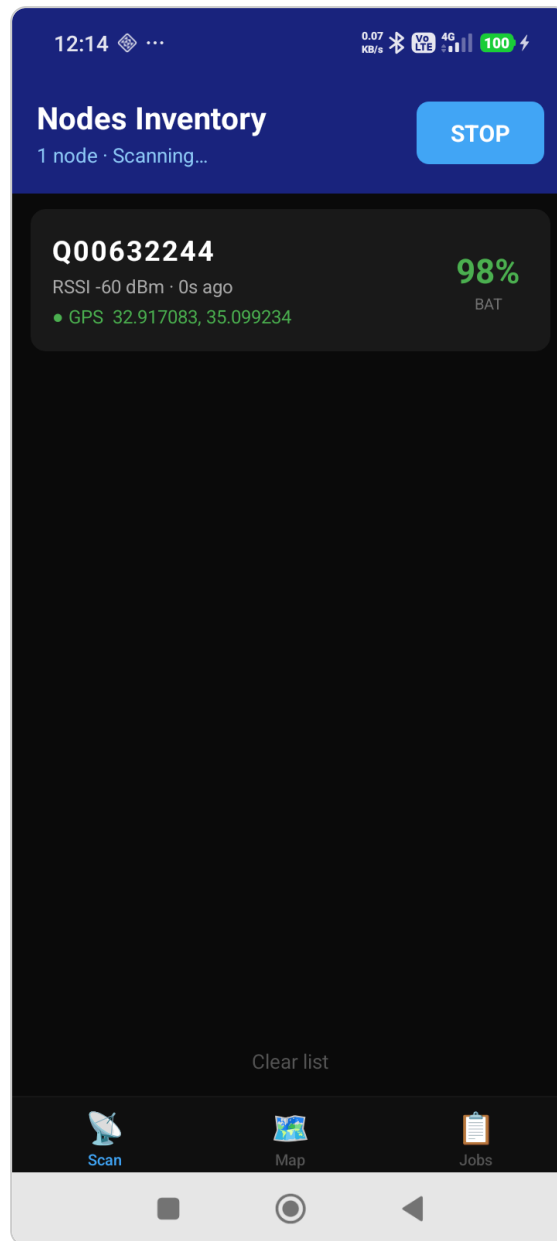
An OpenStreetMap view with a pin for every node that has GPS coordinates. Use it to see where located nodes actually sit.

Jobs

The deploy-and-pickup accounting workflow. This is where a job is created, deployed, picked up, reconciled, reported, and closed.

Scan

The Scan tab is a live look at every node broadcasting near you right now.



The Scan screen showing a detected node

- Press **SCAN** to start listening and **STOP** to stop.
- If Bluetooth is off, an alert asks you to turn it on. Turn Bluetooth on, then press **SCAN** again.
- Each node in range appears as its own card. Nodes show up as they broadcast, so give it a few seconds; they do not all appear at once.
- Press **Clear list** to empty the list and start fresh.

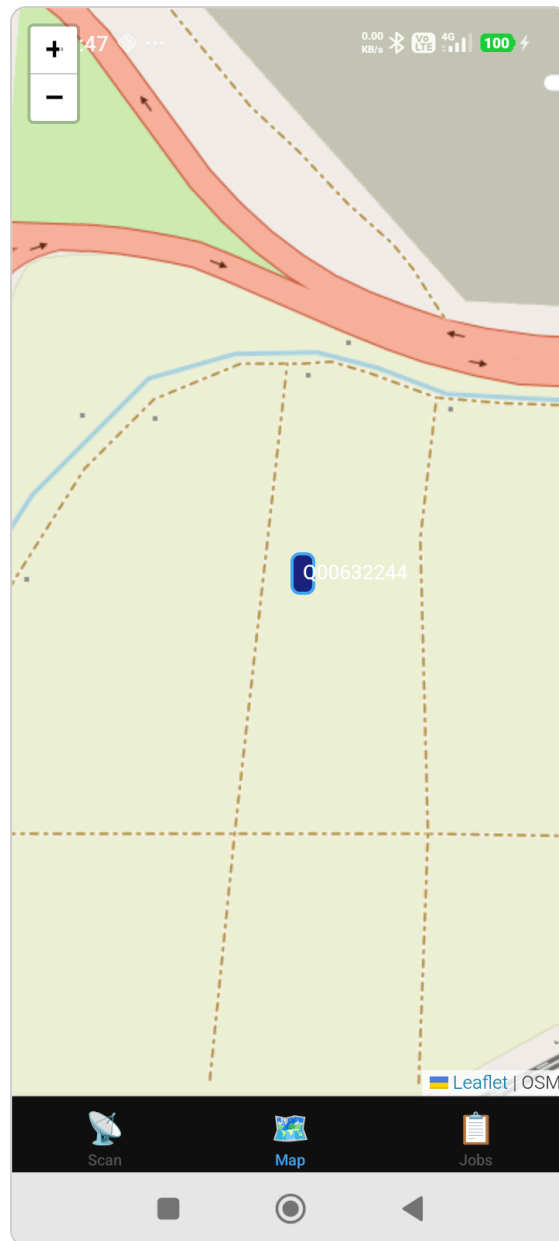
What a node card shows

- **Q-label:** the serial printed on the unit's physical label, for example **Q00599373**.

- **Signal:** a line like "RSSI -59 dBm, 3s ago", meaning the last signal strength and how long ago it was last heard.
- **GPS:** the coordinates when the node has a fix, otherwise `No GPS`.
- **Battery:** a percentage, color-coded: ● green above 50, ● orange above 20, ● red when low.

Map

The Map tab plots your located nodes on an OpenStreetMap background.



The Map with a node pin

- Every node that has GPS coordinates gets a blue pin labeled with its Q-serial.
- Tap a pin to open a popup with battery, RSSI, latitude, longitude, and a [Remove from map](#) button that clears that node.
- If no node has a GPS fix yet, a banner tells you so. Nodes without coordinates do not appear as pins.

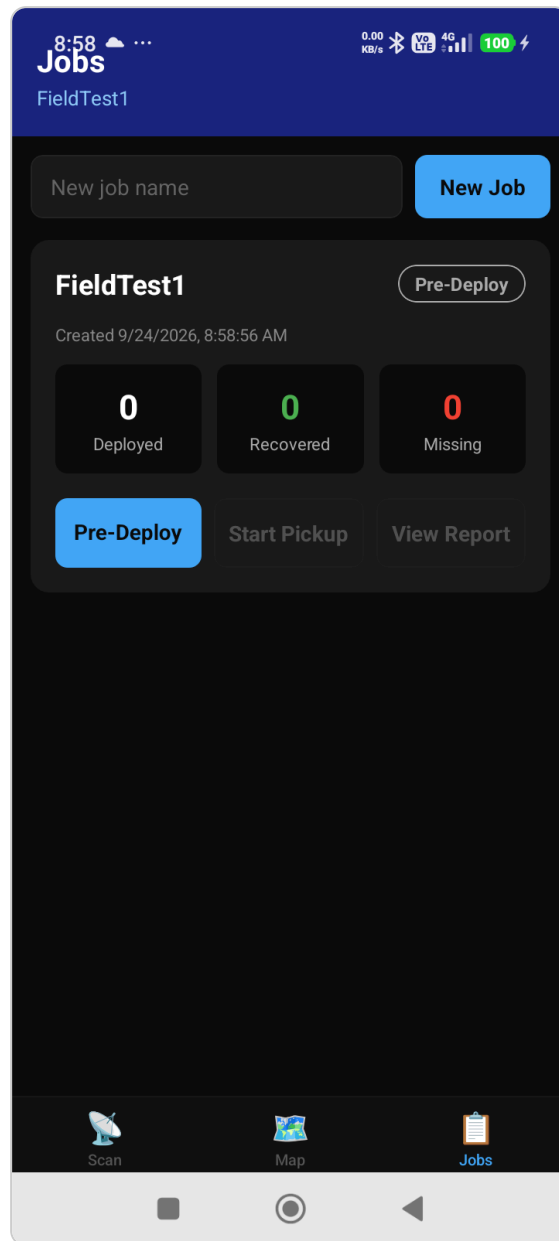
Jobs

A job is one deployment cycle: the nodes you put out, and the accounting that proves you got them back. Every job moves through four statuses.

- **Pre-Deploy:** you are building the manifest by scanning the nodes before they go out.
- **Deployed:** the manifest is saved and the nodes are in the field.
- **Pickup:** you are reconciling the collected nodes against the manifest.
- **Closed:** the job is finished and its report has been saved.

Jobs Home

The Jobs tab opens on the jobs home, where you create and manage jobs.

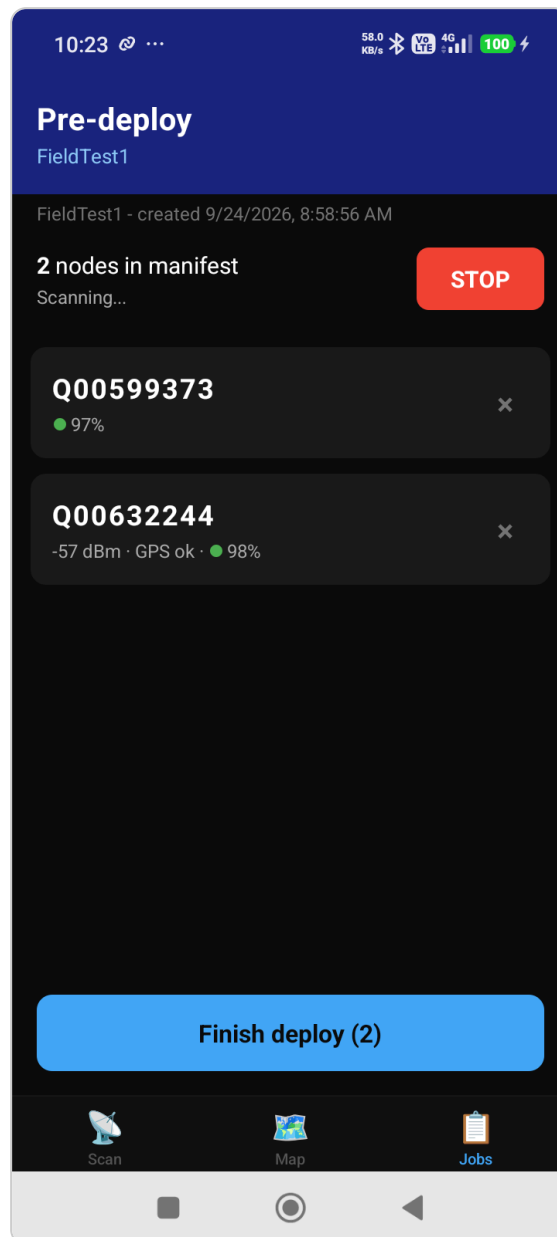


A job with its status and counts

- Type a name and press **New Job** to create a job.
- The active job card shows the name, a status pill, timestamps (Created, Deployed, Picked up), and three counts: **Deployed**, **Recovered**, and **Missing**.
- Its buttons are **Pre-Deploy**, **Start Pickup** (enabled once the job is deployed), and **View Report**.
- Each job has a **Delete** action with a confirm dialog. Deleting is recorded in the log.
- A **History** button opens the operations log.
- Past jobs are listed below the active job.

Pre-Deploy

Pre-Deploy builds the manifest before the nodes go out: the official list of which nodes belong to this job. Do this while all the nodes are still together in their cradles or bags, not out in the field.



Pre-Deploy: the manifest being built, each node with a remove control

1. With the nodes together in their cradles or bags, press **SCAN**. Every node the phone hears is added to this job's manifest with its label, battery, RSSI, and GPS.
2. If a node was scanned by mistake, press the small remove (X) on its row to drop it.
3. Press **Finish deploy (N)** to save the manifest and mark the job Deployed. N is the number of nodes on the manifest.
4. Only then send the nodes out to the field.

TIP

Keep the nodes close together while you scan so each one is heard clearly before you finish. The manifest is your source of truth for pickup, so make sure it lists every node you actually send out.

Pickup

Pickup checks the nodes that came back from the field against the manifest and tells you if any are still missing. Do this once all the nodes are back and sitting together in their bags or cradles.

1. With the returned nodes together in their bags or cradles, press **SCAN**, then **Reconcile now**.
2. The app compares each manifest node against what it just heard over Bluetooth. A node is confirmed by its Bluetooth signal, not by GPS.
3. A node is marked **Recovered** when it is heard at strong signal (strong RSSI), which is the proof it is right there in the bag or cradle next to the phone.
4. Once a node is Recovered it stays Recovered across repeated passes, so you can keep scanning without losing progress.

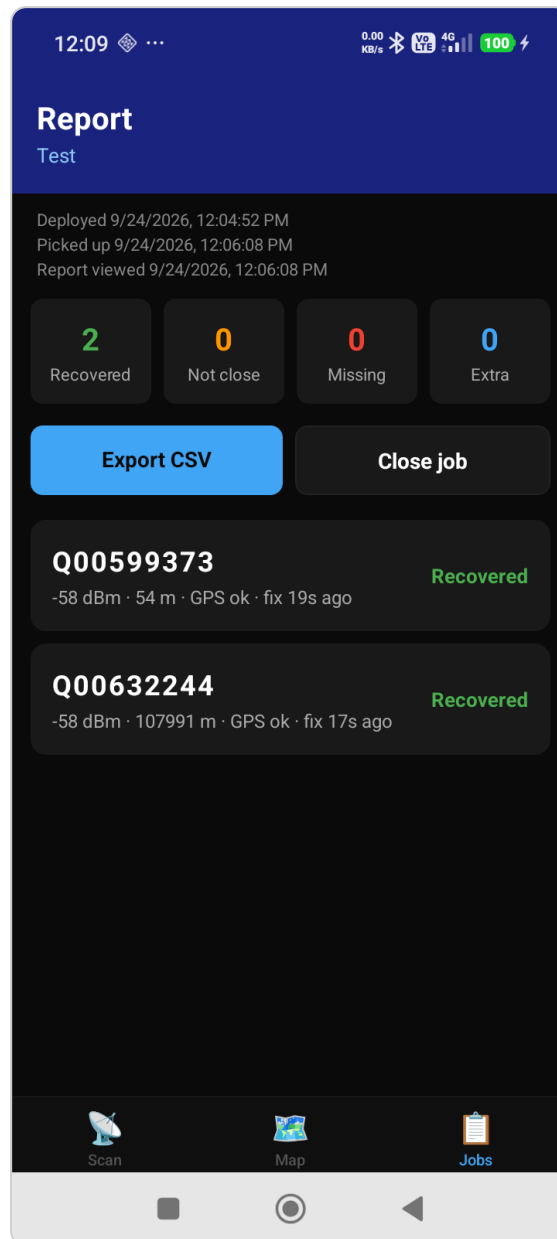
The header shows "R of M recovered" (recovered count out of the manifest count). Each row shows its Bluetooth signal (RSSI) and a status badge, and for reference any position info the node reported (distance and GPS status).

TIP

If some nodes still show as not recovered, bring them closer to the phone and press **Reconcile now** again. Nodes already counted stay counted.

Report

The report is the final tally for the job and the file you hand off.



The Report: counts and per-node results

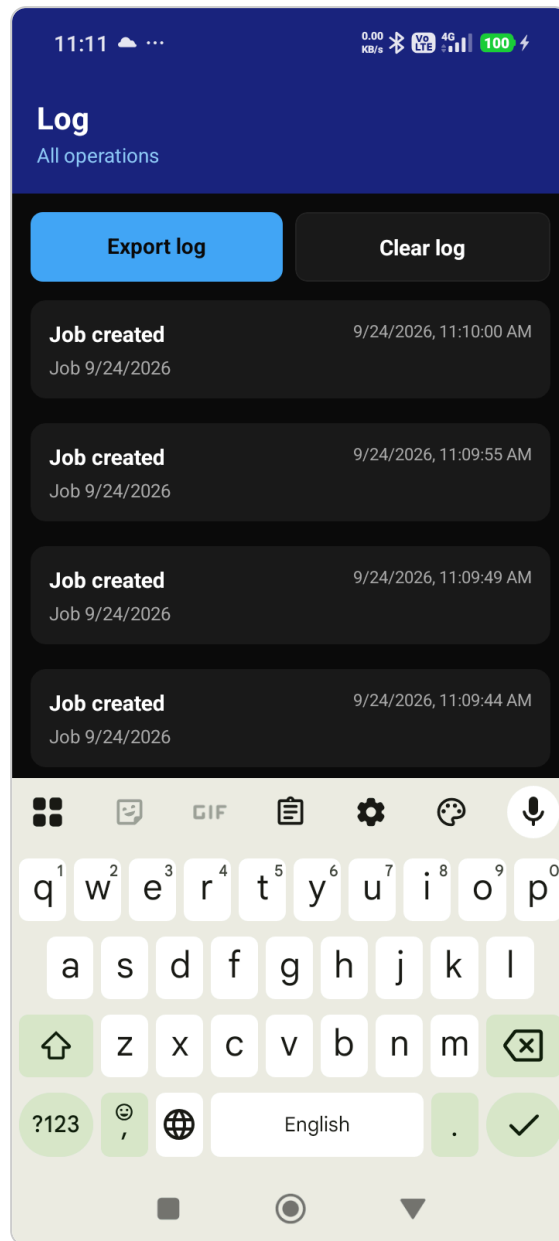
- Four count tiles at the top: **Recovered**, **Not close**, **Missing**, and **Extra**.
- **Export CSV** saves a file named like `JobName_2026-09-24T10-12-24.csv` into device storage and opens the system share sheet so you can send it.
- **Close job** marks the job Closed.
- Below the tiles, every node is listed with its state.

WHAT THE TILES MEAN

Recovered is confirmed in hand or next to the phone. **Not close** was heard but not close enough to confirm. **Missing** was not heard in this pickup pass. **Extra** is a node heard that was not on the manifest.

History (Log)

The history is an append-only record of everything the app did, newest first.



The operations Log

- It records job created, status changes, deploy finished, node removed, reconcile result counts, CSV exports, and job deletions (each with a short summary).
- You can **Export** the log to CSV, or **Clear** it.

NOTE

The log is append-only: entries are added, never edited. It is your audit trail for what happened on each job and when.

Key Concepts

A node is its Q-serial

Every node is identified by its Q-serial, the same code printed on its physical label, for example `Q00599373`. When in doubt, match the label on screen to the label on the unit in your hand.

Where the numbers come from

Battery, GPS, and RSSI all come from the node's own BLE broadcast. The node announces itself over the air and the app simply listens. That is why the values update as you move, and why a node has to be broadcasting to appear at all.

RSSI, in plain terms

RSSI is the signal strength in dBm. Closer is stronger. For example, -55 is strong and -90 is weak. A strong signal means the node is near you.

The recovery rule

Recovered means the node is physically back with you, in its bag or cradle. The strongest proof is a strong Bluetooth signal at close range: since all the nodes sit together with the phone during pickup, a node that made it back is heard strong and clear. This is why a node in the bag shows Recovered even if its stored GPS still reads its old field position: the app trusts the live signal over the old coordinates.

Troubleshooting

Bluetooth is off

If pressing `SCAN` shows an alert, Bluetooth is off. Turn it on, then press `SCAN` again. Nothing can be heard while Bluetooth is off.

No nodes appear

Check that Bluetooth is on and location is enabled, then wait a few seconds. Nodes broadcast periodically, not continuously, so they arrive over the first several seconds of a scan. If a node is far away, move closer.

GPS shows No GPS, or a node is not on the map

A node reads `No GPS` until it gets its own satellite fix, and the map only pins nodes that have coordinates. A GPS fix is not needed for collection. A node is confirmed by its Bluetooth signal and is Recovered even with no GPS at all. GPS only affects the Map and the informational distance and GPS fields shown on a row.

A node shows Missing on the report

Missing means the node was not heard during that pickup pass, so it was not confirmed in the bag or cradle. Check the pile again, press `SCAN`, then `Reconcile now` again. Nodes already recovered stay recovered.

